

Axiom Switch Universe Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Extended	ID	axiom-switch-universe

Big idea: Axioms create mathematical worlds.

Student-Facing Prompt

Inspect Axiom Switch Universe Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Axiom Switch Universe Stimulus A	Student-facing stimulus 1: A geometry rule change where parallel lines behave differently. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Axiom Switch Universe Stimulus B	Student-facing stimulus 2: A made-up arithmetic system with a changed operation rule. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Axiom Switch Universe Reveal / Twist	Student-facing stimulus 3: A reflection on whether certainty depends on starting assumptions. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Axioms create mathematical worlds.

Actual Lesson Questions

#	Question for students
1	After inspecting Axiom Switch Universe Stimulus A, what is your first knowledge claim?
2	Which exact detail from Axiom Switch Universe Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Are mathematical truths absolute?

Ready-to-Use Examples

- A geometry rule change where parallel lines behave differently.
- A made-up arithmetic system with a changed operation rule.
- A reflection on whether certainty depends on starting assumptions.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Give groups different geometry worlds: exactly one parallel line, no parallel lines, or many parallel lines through a point not on a line.
2. Students draw what triangles, lines and maps might be like in their world.
3. Each group states one theorem that may change.
4. Discuss usefulness, consistency and truth.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Are mathematical truths absolute?
- If different axiom systems are possible, what makes one useful?
- Is consistency enough for mathematical knowledge?

Knowledge Questions

- Are mathematical truths absolute?
- If different axiom systems are possible, what makes one useful?
- Is consistency enough for mathematical knowledge?

Transfer Example

For example, run this activity with a geometry rule change where parallel lines behave differently. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...